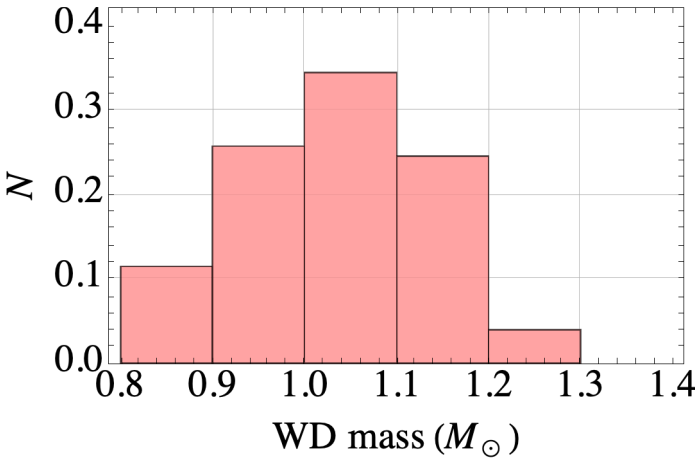
In[1]:=
```
SetDirectory["/Users/lucym/Documents/2026/Maria-visit"]
```
Out[1]=
```
/Users/lucym/Documents/2026/Maria-visit
```

Section 2 (Motivation) : the sample from Cheng et al 2020

Taking the information in the caption of Fig 3. in Cheng + 2020, we can determine the total number of merger remnants in each mass bin.
408 (0.07), 431 (0.15), 323 (0.27), 176 (0.35), and 57 (0.18)
= 29, 65, 87, 62, 10

In[2]:=
```
bin1 = Table[0.85, {i, 1, 29}];
bin2 = Table[0.95, {i, 1, 65}];
bin3 = Table[1.07, {i, 1, 87}];
bin4 = Table[1.19, {i, 1, 62}];
bin5 = Table[1.28, {i, 1, 10}];
```

In[7]:=
```
cheng3 = Histogram[Join[bin1, bin2, bin3, bin4, bin5], 8, "Probability",
  ChartStyle → {Directive[Blend[{Yellow, Magenta}], Opacity[0.7]]},
  Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
  FrameLabel → {Style["WD mass (M⊙)"], Style["N"]}, BaseStyle → {FontSize → 20},
  GridLines → Automatic, PlotRange → {{0.8, 1.4}, {0, 0.4}}]
```
Out[7]=



Section 3.1/3.2 (Results) : Constructing an intrinsic mass distribution for white dwarfs in double white dwarf binaries (DWDBs)

First we take the sample of white dwarfs in binaries collated by James Munday available here : https://github.com/JamesMunday98/CloseDWDbinaries/blob/main/CloseDoubleWhiteDwarfs_20260216.csv
We ultimately seek an intrinsic mass distribution N_{MW}(m) of white dwarfs in short period (potential merger progenitors, periods < 3 hr). From this sample, we seek an observed, roughly magnitude

limited mass distribution N_{obs}(m) with which we can deform according to how much more difficult (easy) it is to observe higher (lower) mass WD for a fixed surface temperature. From this sample, we only select binaries which have both m_1 and m_2 constrained, and at least one of the component temperatures measured.

In[8]:=
```
MundaySample = "munday-cut-4.csv";
m1s = Select[NumericQ][Import[MundaySample, {"Data", All, 2}]];
m2s = Select[NumericQ][Import[MundaySample, {"Data", All, 3}]];
T1s = Select[NumericQ][Import[MundaySample, {"Data", All, 4}]];
T2s = Select[NumericQ][Import[MundaySample, {"Data", All, 5}]];
```

In[13]:=
```
{Length[T1s], Length[m1s], Length[m2s], Length[T2s]}
```

Out[13]=
```
{134, 134, 134, 134}
```

But there are several secondary temperatures which aren' t measured .

In[14]:=
```
T1smod = Pick[T1s, # > -50 & /@ (T1s)];
m1smod = Pick[m1s, # > -50 & /@ (m1s)];
m2smod = Pick[m2s, # > -50 & /@ (m2s)];
T2smod = Pick[T2s, # > -50 & /@ (T2s)];
```

In[18]:=
```
{Length[T1smod], Length[m1smod], Length[m2smod], Length[T2smod]}
```

Out[18]=
```
{134, 134, 134, 72}
```

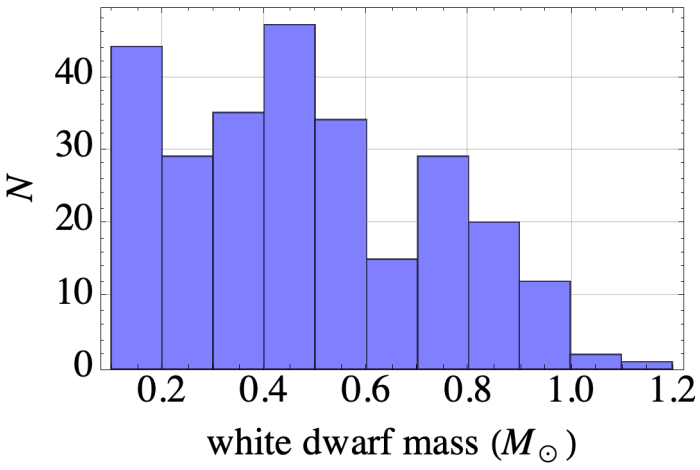
So if we only want the pairs that have both m1 and m2, we can select the corresponding m1' s

In[19]:=
```
m1pair = Pick[m1s, # > -50 & /@ (m2s)];
```

This is the observed mass distribution in these 134 binaries

In[20]:=
```
binaryWDmass =
 Histogram[Join[m1pair, m2smod], 10, ChartStyle → {Blend[{Magenta, Cyan}]},
  Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
  FrameLabel → {Style["white dwarf mass (M⊙)"], Style["N"]},
  BaseStyle → {FontSize → 20}, GridLines → Automatic]
```

Out[20]=

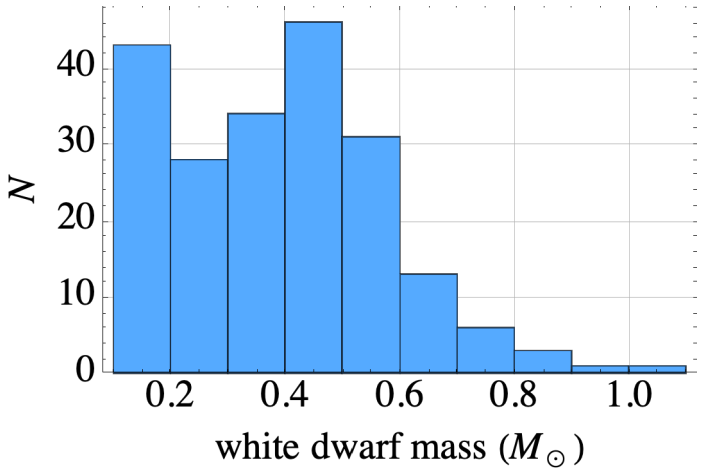


But we will use only the components that have temperature measurements (blue)

In[21]:=
```
m1temps = Pick[m1s, # > -50 & /@ (T1s)];
m2temps = Pick[m2s, # > -50 & /@ (T2s)];
m2tempsfix = Pick[m2temps, # > -50 & /@ (m2temps)];
```

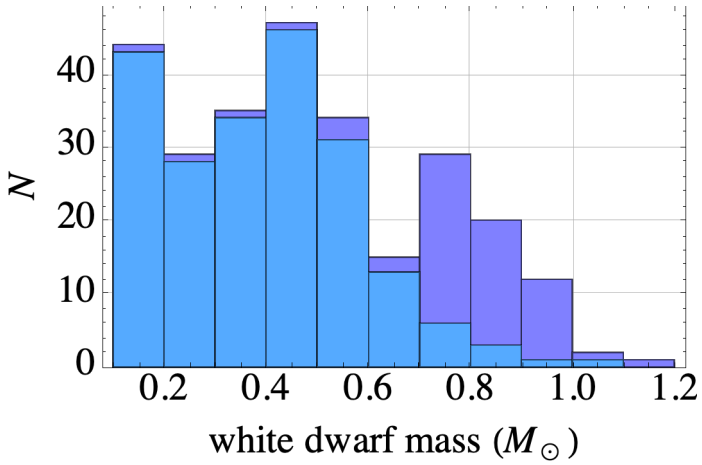
In[24]:=
```
binarytempmass = Histogram[Join[m1temps, m2tempsfix],
  10, ChartStyle → {Blend[{Magenta, Cyan, Cyan}]},
  Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
  FrameLabel → {Style["white dwarf mass (M⊙)"], Style["N"]},
  BaseStyle → {FontSize → 20}, GridLines → Automatic]
```

Out[24]=



In[25]:=
```
Show[binaryWDmass, binarytempmass]
```

Out[25]=



In[26]:=
```
T1temps = Pick[T1s, # > -50 & /@ (T1s)];
T2temps = Pick[T2s, # > -50 & /@ (T2s)];
T2tempsfix = Pick[T2temps, # > -50 & /@ (m2temps)];
```

In[29]:=
```
bluemass = Join[m1temps, m2tempsfix];
blueTemps = Join[T1temps, T2tempsfix];
{Length[T1temps], Length[T2tempsfix]}
```

Out[31]=
```
{134, 72}
```

So there are 206 measurements of mass and temperatures of white dwarfs in binaries.

Selection function

We have analytic fits to both He and CO core white dwarfs . These are (in units of solar mass)

In[32]:=
```
RHe[m_, T_] := 0.0132 × 10^(-0.177 T^0.5) m^(0.148-0.941 T^0.5)
RCO[m_, T_] := 0.9 × 10^(-0.00063 T^2) m^(-0.5-0.13 T) / 100
```

In[34]:=

Now we approximate the radius of each white dwarf, assuming that all masses < 0.45 and He core and all > 0.45 are CO core.

In[35]:=
```
HeCObmass = 0.45;
```

In[36]:=
```
HeWDm = Pick[bluemass , # < HeCObmass & /@ (bluemass )];
HeWDT = Pick[blueTemps , # < HeCObmass & /@ (bluemass )];
COWDm = Pick[bluemass , # ≥ HeCObmass & /@ (bluemass )];
COWDT = Pick[blueTemps, # ≥ HeCObmass & /@ (bluemass )];
```

The mean temperature and mass are :

In[40]:=
```
reftemp = Mean[HeWDT] 10^-4 // N
refmass = Mean[HeWDm] // N
```

Out[40]=
```
1.35209
```

Out[41]=
```
0.27748
```

Selection function Step 1/2 : assign less weight to exceptionally hot white dwarfs by calculating the ratio of white dwarf luminosity to its reference luminosity, if it had the typical temperature of the sample

Their approximate luminosities

In[42]:=
```
HeL = RHe[HeWDm, HeWDT 10^-4]^2 HeWDT^4;
COL = RCO[COWDm, COWDT 10^-4]^2 COWDT^4;
```

Their reference luminosities

In[44]:=
```
HeLref = RHe[HeWDm, reftemp]^2 HeWDT^4;
COLref = RCO[COWDm, reftemp]^2 COWDT^4;
```

Selection function Step 2/2 : This is how much we should repeat each mass by, relative to how much more difficult / easy it is to see than a 0.2 solar mass WD .

In[46]:=
```
L20ref = RHe[refmass, reftemp]^2 (reftemp 10^4)^4;
```

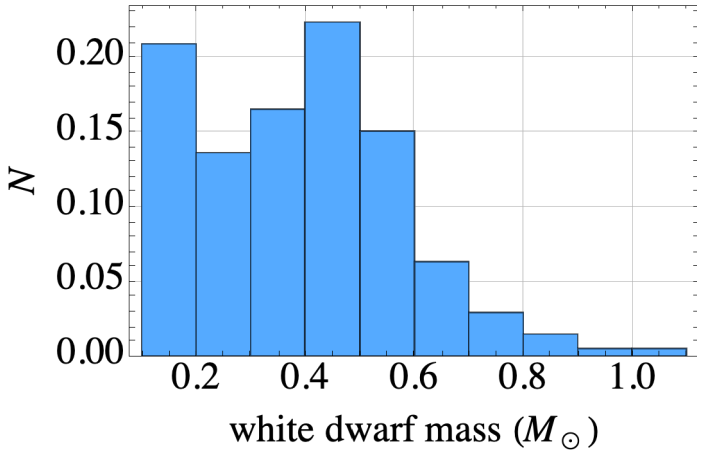
In[47]:=
```
HeLsize = L20ref / (RHe[HeWDm, reftemp]^2 (reftemp 10^4)^4);
COLsize = L20ref / (RCO[COWDm, reftemp]^2 (reftemp 10^4)^4);
```

In[49]:=
```
repeatHe = 1000 (HeLref / HeL) HeLsize ;
repeatCO = 1000 (COLref / COL) COLsize;
```

In[51]:=
```
Hescaled =
  Table[Table[HeWDm[[i]], {j, 1, repeatHe[[i]]}], {i, 1, Length[HeWDm]}] // Flatten;
COscaled =
  Table[Table[COWDm[[i]], {j, 1, repeatCO[[i]]}], {i, 1, Length[COWDm]}] // Flatten;
```

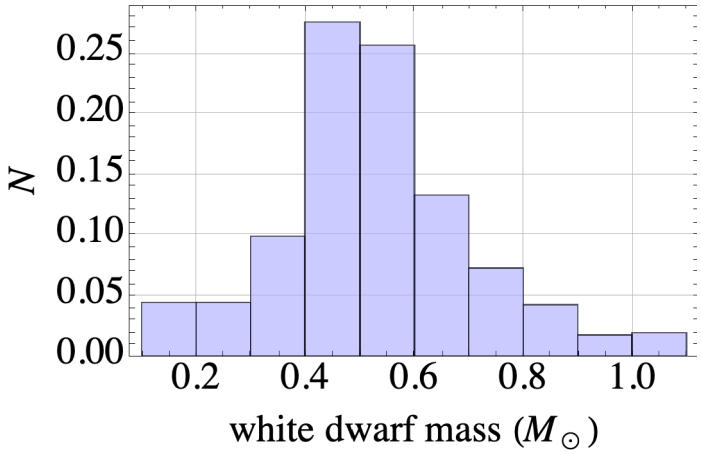
In[53]:=
```
binarytempmass2 = Histogram[Join[m1temps, m2tempsfix],
  10, "Probability", ChartStyle → {Blend[{Magenta, Cyan, Cyan}]},
  Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
  FrameLabel → {Style["white dwarf mass (M⊙)"], Style["N"]},
  BaseStyle → {FontSize → 20}, GridLines → Automatic]
```

Out[53]=



In[54]:=
```
correctedmasses = Histogram[Join[Hescaled, COscaled], 10, "Probability",
  ChartStyle → {Directive[Blend[{Magenta, Cyan}], Opacity[0.4]]},
  Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
  FrameLabel → {Style["white dwarf mass (M⊙)"], Style["N"]},
  BaseStyle → {FontSize → 20}, GridLines → Automatic]
```

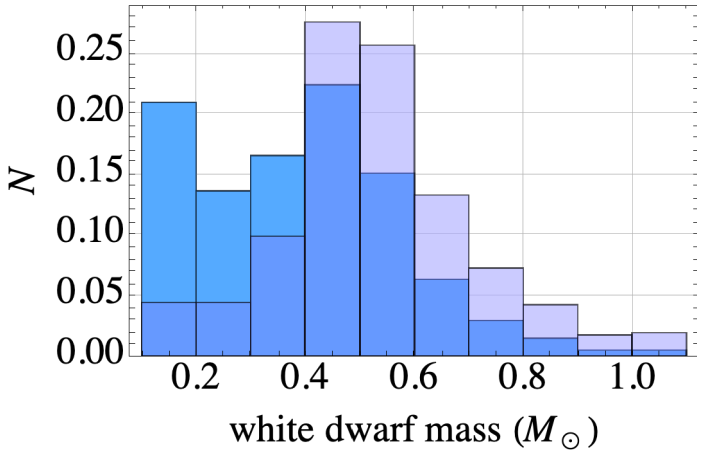
Out[54]=



Comparison with intrinsic (purple) and observed (blue)

In[55]:=
```
Show[binarytempmass2, correctedmasses]
```

Out[55]=



Sampling m2 and m2 from this distribution

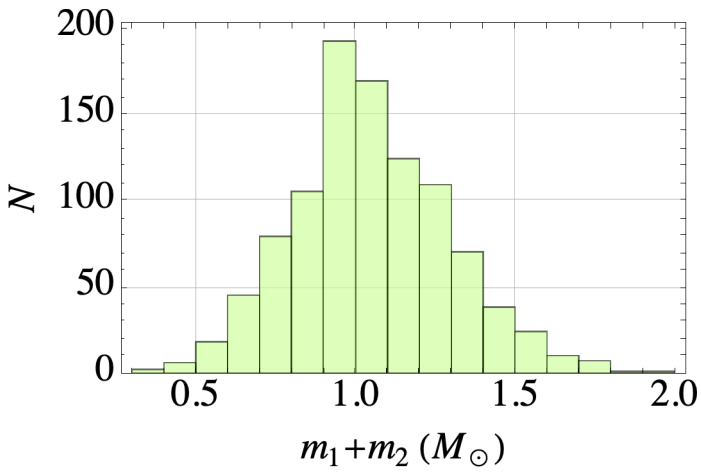
In[56]:=
```
nWD = 1000;
InvCDF[u_] = InverseCDF[HistogramDistribution[Join[Hescaled, COscaled]], u];
```

In[58]:=
```
SeedRandom[];
drawtab = Table[InvCDF[RandomReal[]], 2, {nWD}];
draws = drawtab[[1]];
draws2 = drawtab[[2]];
```

Here are the sums of the draws :

In[62]:=
```
Histogram[ draws + draws2, 20,
 ChartStyle → Directive[{Blend[{Yellow, Yellow, Cyan}]}, Opacity[0.4]],
 Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
 FrameLabel → {Style["m1+m2 (M⊙)"], Style["N"]},
 BaseStyle → {FontSize → 20}, GridLines → Automatic]
```

Out[62]=



In[63]:=

Separating in terms of composition

Cheng' s sample should not include He + He composition mergers, so we should exclude these from our sum sample . First we should assign m1 to be the more massive and m2 to be the less massive from each pairs

In[64]:=
```
m1M = Table[Max[draws[[ii]], draws2[[ii]]], {ii, 1, Length[draws]}];
m2M = Table[Min[draws[[ii]], draws2[[ii]]], {ii, 1, Length[draws]}];
```

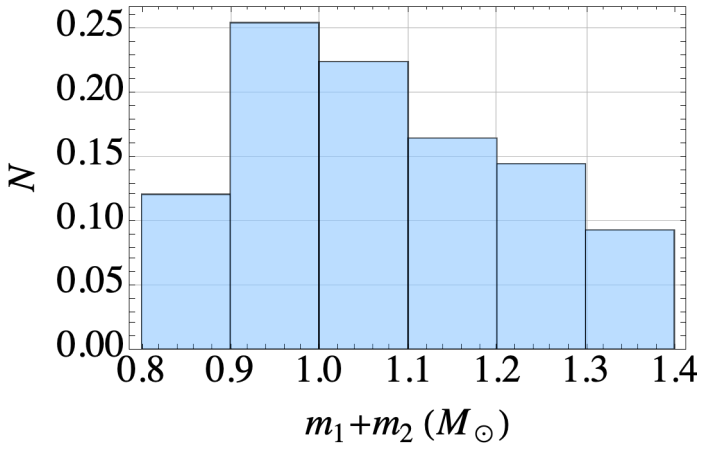
In[66]:=
```
HeCO1 = Pick[m1M, # < HeCObmass & /@ (m1M)];
HeCO2 = Pick[m2M, # < HeCObmass & /@ (m1M)];
HeCO3 = Pick[HeCO2, # > HeCObmass & /@ (HeCO2)];
HeCO4 = Pick[HeCO1, # > HeCObmass & /@ (HeCO2)];
COHe1 = Pick[m1M, # > HeCObmass & /@ (m1M)];
COHe2 = Pick[m2M, # > HeCObmass & /@ (m1M)];
COHe3 = Pick[COHe2, # < HeCObmass & /@ (COHe2)];
COHe4 = Pick[COHe1, # < HeCObmass & /@ (COHe2)];
COCO1 = Pick[m1M, # > HeCObmass & /@ (m1M)];
COCO2 = Pick[m2M, # > HeCObmass & /@ (m1M)];
COCO3 = Pick[COCO1, # > HeCObmass & /@ (COCO2)];
COCO4 = Pick[COCO2, # > HeCObmass & /@ (COCO2)];
HeHe1 = Pick[m1M, # < HeCObmass & /@ (m1M)];
HeHe2 = Pick[m2M, # < HeCObmass & /@ (m1M)];
HeHe3 = Pick[HeHe1, # < HeCObmass & /@ (HeHe2)];
HeHe4 = Pick[HeHe2, # < HeCObmass & /@ (HeHe2)];
```

In[82]:=
```
noHeHe = Join[HeCO3 + HeCO4, COHe3 + COHe4, COCO3 + COCO4];
```

Section 3.3 (Results) : Comparison with Cheng et al 2020

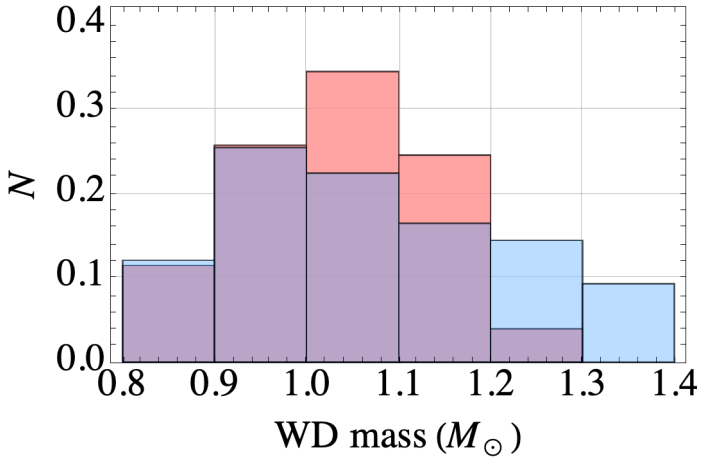
In[83]:=
```
sumguessCOCO3 =
 Histogram[Pick[noHeHe, 0.8 < # < 1.4 & /@ (noHeHe)], 5, "Probability",
  ChartStyle → Directive[{Blend[{Magenta, Cyan, Cyan}]}, Opacity[0.4]],
  Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
  FrameLabel → {Style["m1+m2 (M⊙)"], Style["N"]},
  BaseStyle → {FontSize → 20}, GridLines → Automatic]
```

Out[83]=



In[84]:=
```
Show[cheng3, sumguessCOCO3]
```

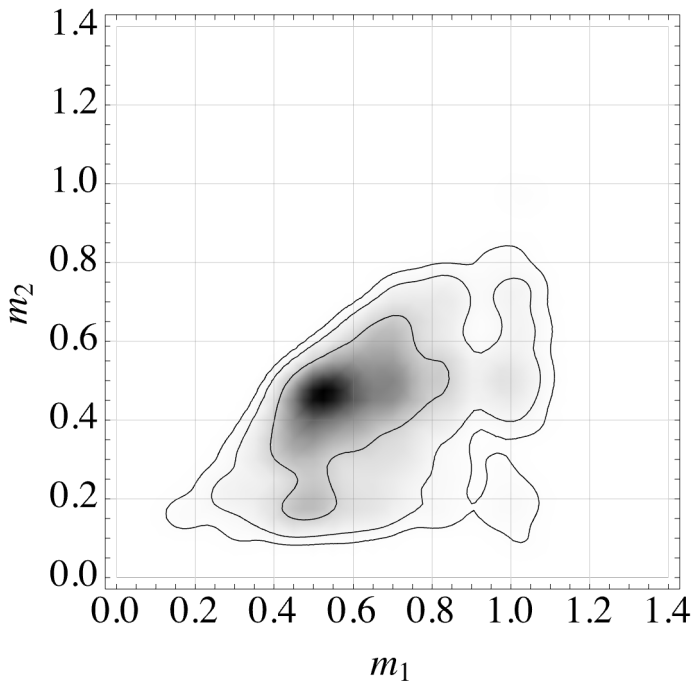
Out[84]=



Section 4 : pairings for m1 and m2

In[85]:=
```
data = Transpose[{m1M, m2M}];
d = SmoothKernelDistribution[data];
pdf = Reverse[
   Sort[Flatten[Table[PDF[d, {x, y}], {x, 0, 1.4, 0.005}, {y, 0, 1.4, 0.005}]]]];
cdf = Accumulate[pdf] / Total[pdf];
contours = pdf[[Flatten[
    Table[FirstPosition[cdf, p_ /; p ≥ alpha], {alpha, {0.68, 0.95, 0.99}}]]]];
contours1 = pdf[[Flatten[
    Table[FirstPosition[cdf, p_ /; p ≥ alpha], {alpha, {0.50, 0.75, 0.90}}]]]];
c = contours;
cPad = PadLeft[c, Length@c + 1, 1];
cPad = PadRight[cPad, Length@cPad + 1, 0];
quantileBounds = Partition[cPad, 2, 1];
splitUp =
  Table[Select[data, Last[i] < PDF[d, #] ≤ First[i] &], {i, quantileBounds}];
sdh3 = SmoothDensityHistogram[data, PlotRange → {{0 × 0.1, 1.4}, {0 × 0.1, 1.4}},
   AspectRatio → 1, Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
   FrameLabel → {Style["m1"], Style["m2"]}, BaseStyle → {FontSize → 20},
   GridLines → Automatic, ColorFunction → (Blend[{White, Black}, #] &)];
cp = ContourPlot[PDF[d, {x, y}], {x, 0, 1.4}, {y, 0, 1.4},
   Contours → contours, ContourShading → None, PlotRange → All];
corner1 = Show[{sdh3, cp}]
```

Out[98]=

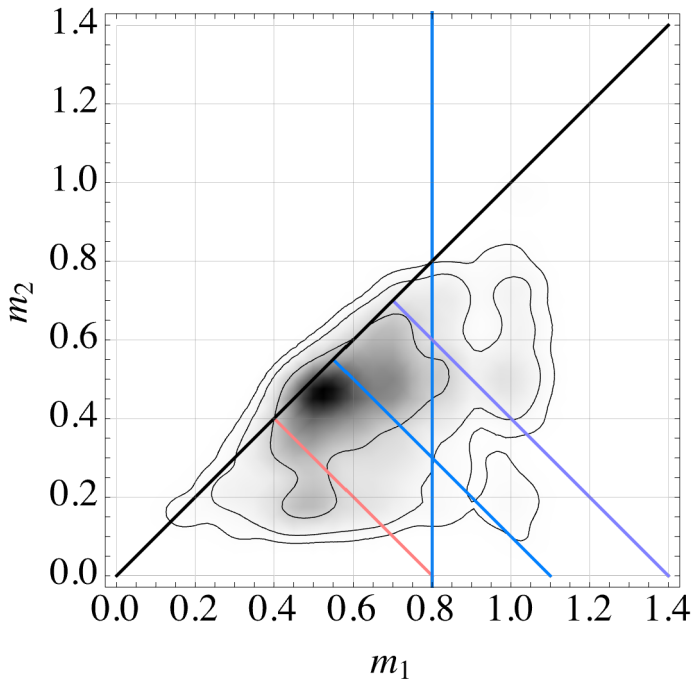


Putting some sums for comparison with Cheng

In[99]:=
```
Ultralimit = Plot[0.8 - x, {x, 0.4, 0.8}, PlotStyle → Blend[{Magenta, Yellow}]];
sublimit = Plot[1.1 - x, {x, 0.55, 1.1}, PlotStyle → Blend[{Cyan, Blue}]];
Ialimit = Plot[10 000 (0.8 - x), {x, 0.5, 1}, PlotStyle → Blend[{Cyan, Blue}]];
Chandralimit = Plot[1.4 - x, {x, 0.7, 1.4}, PlotStyle → Blend[{Cyan, Magenta}]];
equalmass = Plot[x, {x, 0, 1.4}, PlotStyle → Black,
   PlotRange → {{0, 1.4}, {0, 1.4}}, AspectRatio → 1];
```

In[104]:=
```
Show[corner1, Ultralimit, Ialimit, sublimit, Chandralimit, equalmass]
```

Out[104]=



Separations into total mass sum

In[105]:=
```
Mid1 = Pick[m1M, # < 1.4 & /@ (m1M + m2M)];
Mid2 = Pick[m2M, # < 1.4 & /@ (m1M + m2M)];
Masssummid = Mid1 + Mid2;
Mid1a = Pick[Mid1, # > 0.8 & /@ (Masssummid)];
Mid2a = Pick[Mid2, # > 0.8 & /@ (Masssummid)];
```

Sums > 1.4

In[110]:=
```
superC = 1 - Length[Mid1] / Length[(m1M + m2M)] // N
```

Out[110]=
```
0.081
```

Sums < 0.8

In[111]:=
```
ELMmerger = 1 - Length[Mid1a] / Length[(m1M + m2M)] - superC // N
```

Out[111]=
```
0.15
```

The ones between 0.8 - 1.4

In[112]:=
```
subC = 1 - superC - ELMmerger
```
Out[112]=
```
0.769
```

(Just to double check)

In[113]:=
```
Length[Mid1a] / Length[(m1M + m2M)] // N
```
Out[113]=
```
0.769
```

Of the ones between 0.8 - 1.4m we want to see which are above/below 1.1 - which we take as the boundary for a possible sub Chandrasekhar Ia.

In[114]:=
```
Mid1b = Pick[m1M, # < 1.4 & /@ (m1M + m2M)];
Mid2b = Pick[m2M, # < 1.4 & /@ (m1M + m2M)];
Masssummidb = Mid1b + Mid2b;
Mid1c = Pick[Mid1b, # > 1.1 & /@ (Masssummidb)];
Mid2c = Pick[Mid2b, # > 1.1 & /@ (Masssummidb)];
```

The ones between 0.8 - 1.1

In[119]:=
```
subCexplode = Length[Mid2c] / Length[m1M] // N
```
Out[119]=
```
0.303
```

In[120]:=
```
Mid1d = Pick[m1M, # < 1.1 & /@ (m1M + m2M)];
Mid2d = Pick[m2M, # < 1.1 & /@ (m1M + m2M)];
Masssummidd = Mid1d + Mid2d;
Mid1e = Pick[Mid1d, # > 0.8 & /@ (Masssummidd)];
Mid2e = Pick[Mid2d, # > 0.8 & /@ (Masssummidd)];
```

The ones between 1.1 - 1.4

In[125]:=
```
subCnoexplode = Length[Mid2e] / Length[m1M] // N
```
Out[125]=
```
0.466
```

Separations into HeHe, HeCO and COCO

The HeHe ones :

In[126]:=
```
Length[HeHe4] / Length[m1M] // N
```
Out[126]=
```
0.08
```

The HeCO ones :

In[127]:=
```
Length[Join[HeCO4, COHe4]] / Length[m1M] // N
```
Out[127]=
```
0.4
```

The COCO ones :

In[128]:=
```
Length[COCO4] / Length[m1M] // N
```
Out[128]=
```
0.52
```

In[129]:=
```
Msol = 2 × 10^33;
Rsol = 6.955 × 10^10;
G = 6.67 × 10^-8;
c = 3 × 10^10;
```

Some definitions

Eggleton' s zero - temperature mass–radius relation, quoted by Verbunt & Rappaport (1988), or Eq. 24:

In[133]:=
```
Rcold[m11_] := 0.0114 ((m11 / 1.44)^-2/3 - (m11 / 1.44)^2/3)^1/2
    (1 + 3.5 (m11 / (5.7 × 10^-4))^-2/3 + ((5.7 × 10^-4) / m11))^-2/3;
```

The derivative of the Roche radius in the small q approximation for Paczynski (1971) is just 1/3.

In[134]:=
```
ζRL = 1 / 3;
```

The full derivative of the Roche radius in approximation of Eggleton (1983), which we will use, is given by

In[135]:=
$$\zeta RLlist[m11_, m22_] = \frac{3}{2} \times 3^{1/3}\, m22^{2/3}\, (m11 + m22)^{1/3} \left(-\frac{2\, m22^{1/3}}{9 \times 3^{1/3}\, (m11 + m22)^{4/3}} + \frac{2}{9 \times 3^{1/3}\, m22^{2/3}\, (m11 + m22)^{1/3}} \right);$$

From polytropic mass radius relation, Soberman 1997 etc. the primary radius changes with mass loss according to:

In[136]:=
```
ζ2cold = -1 / 3;
```

In[137]:=
```
m1arr = Table[m11, {m11, 0.0001, 1.4, 0.01}] // Flatten;
```

In[138]:=
```
m1list = m1arr;
```

In[139]:=
$$JJ[m11_, m22_, R_] = -\frac{32}{5} \frac{G^3}{c^5} \frac{m11\, m22\, (m11 + m22)}{\left(\frac{R}{0.46} \left(\frac{m11 + m22}{m22}\right)^{1/3}\right)^4};$$

In[140]:=
```
Medd22[m11_] := 5 × 4.4 × 10^-6 (m11)^2 / (π 10^7) Msol
```

In[141]:=
```
Mdotacc[m11_, m22_] := JJ[m11, m22, Rcold[m22 / Msol] Rsol]
  m22 (1 + ζ2cold / 2 - ζRLlist[m11, m22] / 2 - m22 / m11)^-1
```

In[142]:=
```
bounda = m22 /. Table[FindRoot[Mdotacc[Msol m11, Msol m22] == -Medd22[m11],
      {m22, 1 / 5 m11}], {m11, 0.0001, 1.4, 0.01}] // Flatten;
```

FindRoot: The line search decreased the step size to within tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the merit function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindRoot: The line search decreased the step size to within tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the merit function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindRoot: The line search decreased the step size to within tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the merit function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindRoot::lstol will be suppressed during this calculation.

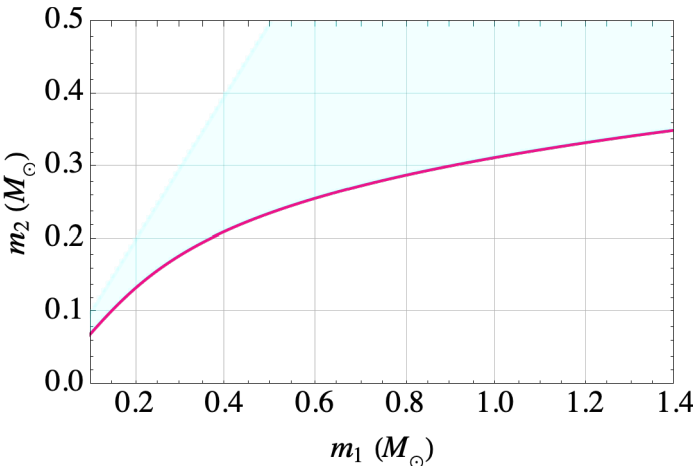
In[143]:=
```
m2up = m22 /. Table[FindRoot[m11 == m22, {m22, 1 / 5 m11}], {m11, 0.0001, 1.4, 0.01}];
```

In[144]:=
```
l1 = Transpose[{m1arr, m2up}];
```

In[145]:=
```
linea = ListPlot[Transpose[{m1arr, bounda}], Joined → True,
    PlotStyle → {Blend[{Magenta, Red}], Thickness[0.005]},
    Frame → True, PlotRange → {{0.1, 1.4}, {0, 0.5}},
    LabelStyle → {(FontFamily → "Times"), Black},
    FrameLabel → {Style["m₁ (M⊙)", 16], Style["m₂ (M⊙)", 16]},
    BaseStyle → {FontSize → 16}, GridLines → Automatic];
l2 = Transpose[{m1arr, bounda}];
fill2 = ListPlot[{l1, l2}, Mesh → All,
    PlotMarkers → None, Joined → True, Filling → {1 → {2}},
    FillingStyle → {Cyan, Opacity[0.05]}, PlotStyle → {{Cyan, Opacity[0.05]}}];
cText = Graphics[Text["merger", {1.1, 0.9}]];
```

In[149]:=
```
Show[linea, fill2, cText]
```

Out[149]=



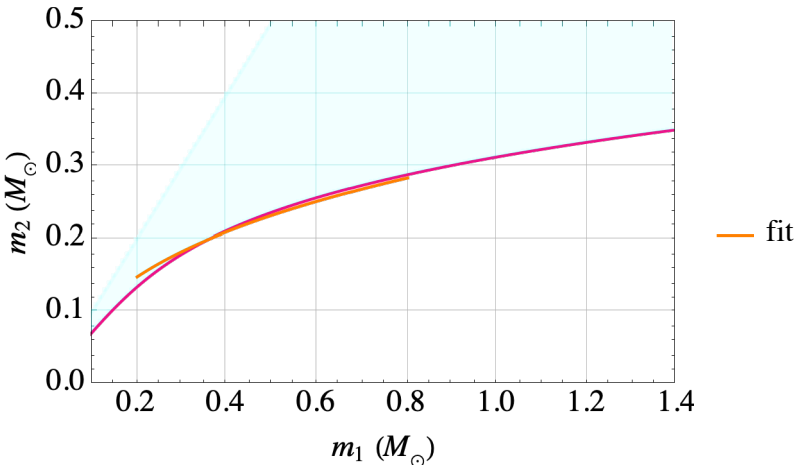
In[150]:=
```
model2 = LinearModelFit[Transpose[{m1arr, bounda}], x^1/4, x]
```
Out[150]=
```
FittedModel[ -0.178711 + 0.490725 x^1/4 ]
```

Separation into unstable and stable mass transfer

The following function is an approximation for the separations between stable and unstable mass transfer according to Nelemans + 2001

In[151]:=
```
fit2 = Plot[-0.17871147963385745` + 0.49072476542885` m^1/4, {m, 0.2, 0.8},
  AspectRatio → 1, AxesLabel → {Style["m1", 16], Style["m2", 16]},
  BaseStyle → {FontSize → 15}, LabelStyle → (FontFamily → "Times"),
  PlotStyle → Blend[{Orange, Orange, Orange}],
  PlotRange → {{0, 0.81}, {0, 0.8}}, PlotLegends → {Style["fit", 16]},
  FrameStyle → Gray, Frame → True, GridLines → Automatic];
```

In[152]:=
```
Show[linea, fill2, cText, fit2]
```
Out[152]=



Unstable vs stable in the 0.8-1.4

In[153]:=
```
boundaryval = -0.17871147963385745` + 0.49072476542885` Mid1a^1/4;
```

In[154]:=
```
Pick[Mid2a, # < 0 & /@ (boundaryval - Mid2a)] // Length
```
Out[154]=
```
730
```

In[155]:=
```
Pick[Mid2a, # > 0 & /@ (boundaryval - Mid2a)] // Length
```
Out[155]=
```
39
```

Unstable vs stable for <0.8

In[156]:=
```
Mid1f = Pick[Mid1, # < 0.8 & /@ (Masssummid)];
Mid2f = Pick[Mid2, # < 0.8 & /@ (Masssummid)];
```

In[158]:=
```
boundaryvalHeHe = -0.17871147963385745` + 0.49072476542885` Mid1f^(1/4);
```

In[159]:=

In[160]:=
```
Pick[Mid2f, # < 0 & /@ (boundaryvalHeHe - Mid2f)] // Length
```

Out[160]=
```
75
```

In[161]:=
```
Pick[Mid2f, # > 0 & /@ (boundaryvalHeHe - Mid2f)] // Length
```

Out[161]=
```
75
```

about half are stable vs unstable

All of them

In[162]:=
```
boundaryvalall = -0.17871147963385745` + 0.49072476542885` m1M^(1/4);
```

In[163]:=
```
Pick[m2M, # < 0 & /@ (boundaryvalall - m2M)] // Length
```

Out[163]=
```
886
```

In[164]:=
```
Pick[m2M, # > 0 & /@ (boundaryvalall - m2M)] // Length
```

Out[164]=
```
114
```

around 10 % are stable according to Nelemans + 2001

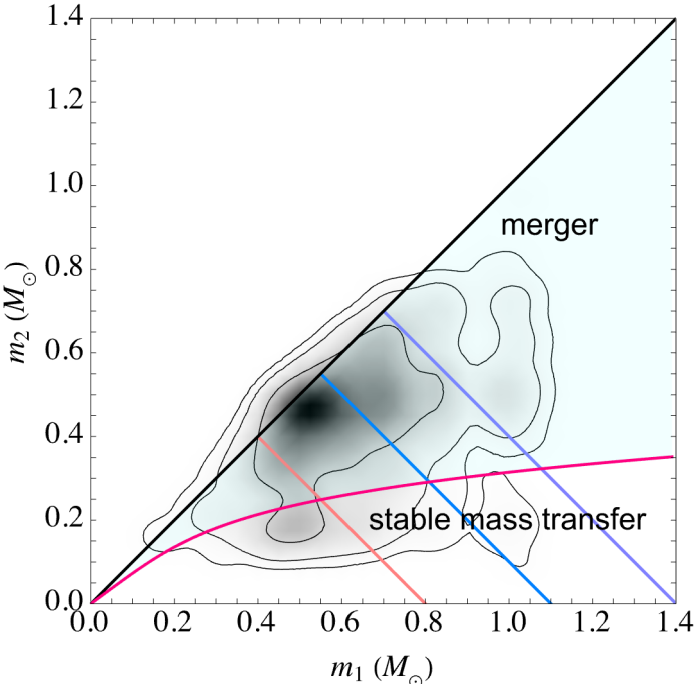
In[165]:=
```
dText = Graphics[Text["stable mass transfer", {1, 0.2}]];
```

Figure 2

In[166]:=
```
lineb = ListPlot[Transpose[{m1list, m1list}], Joined → True,
   PlotStyle → {Black, Thickness[0.005]}, AspectRatio → 1, Frame → True,
   PlotRange → {{0, 1.4}, {0, 1.4}}, LabelStyle → {(FontFamily → "Times"), Black},
   FrameLabel → {Style["m₁ (M⊙)", 16], Style["m₂ (M⊙)", 16]},
   BaseStyle → {FontSize → 16}, GridLines → Automatic];
```

In[167]:=
```
Show[lineb, corner1, equalmass, corner1, Ultralimit,
 sublimit, Chandralimit, fill2, equalmass, linea, cText, dText]
```

Out[167]=



Section 3 : Predictions for the types of binaries in the 100pc sample (and galaxy) (Fig 4)

There are about 100 white dwarf binaries in the Munday sample with periods < 333 min (i . e . 0.1 mHz, possible Hubble time mergers). They are enclosed within 1-2 kpc.
Like we discuss in McNeill and Hirai 2025, focussing on the eclipsing shorter period ones (where selection effects are minor) can give a lower bound on the rate at which double white dwarfs (containing a He component) merge. This is similar to what has been done with the ELM survey etc.
So we can plot the distribution of observed binaries against what we predict in the intrinsic population.
The integral underneath gives you the total number of intrinsic binaries.
Comparing with the observed period distribution, we compute the fraction observed as a function of period (kind of a proxy for R/a).

Using this plot, and the predictions for the various merger rates in our table, we can estimate roughly how many hubble time Ia progenitors are hiding in the 100 pc sample to be detected by e.g. LISA or more sophisticated spectroscopic searches.

Finally we should comment on how these compare to what population synthesis predicts (note that Galactic bulge should have a lot more..)
Hopefully within a factor of 10 or so.

The chirp mass for each combination are roughly :

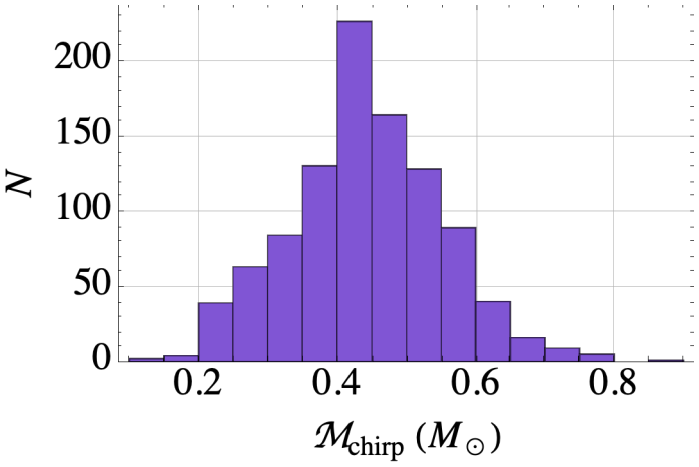
In[168]:=

```
Mchirpf[m11_, m22_] = (m11 m22)^(3/5) / (m11 + m22)^(1/5);
```

In[169]:=

```
Histogram[Mchirpf[draws, draws2], ChartStyle → {Blend[{Purple, Cyan, Magenta}]},
 Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
 FrameLabel → {Style["ℳchirp (M⊙)"], Style["N"]},
 BaseStyle → {FontSize → 20}, GridLines → Automatic]
```

Out[169]=



HeHe

In[170]:=

```
Mchirpf[0.3, 0.3]
```

Out[170]=

```
0.261165
```

HeCO

In[171]:=

```
Mchirpf[0.3, 0.8]
```

Out[171]=

```
0.416724
```

COCO

In[172]:=

```
Mchirpf[0.6, 0.6]
```

Out[172]=

```
0.52233
```

COCO (Ia)

In[173]:=

```
Mchirpf[0.6, 0.8]
```

Out[173]=

```
0.601893
```

For a fixed Mchirp, the number scales with ($\mathcal{M}_{chirp}$)^(-5/3). I.e bigger chirp mass will have less. Because the lower mass ones are hanging around longer. So naturally the COCO ones are probably a few times less frequent for a given frequency range if they have the same merger rate as the HeHe ones. This is broadly consistent with population synthesis.

Cheng+2020 determined that the rate of mergers which sum to 1.1-1.4 solar mass is ~0.002 per year (R Crb rate). However, we predict that this is underestimated by at least half.

So this bin 1.1-1.4 is 0.004 per year.
The fraction between 0.8-1.1 was roughly the same, 0.004 per year.
The HeHe rate was only about half of this, 0.002.
The super Chandra COCO rate was 0.001.

Note that stable mass transfer is predicted to happen about 10% of the time for HeCO, and half the time for HeHe.

OK, now we have the rates which drive mergers. Let's calculate the intrinsic number.

In[174]:=
```
dfdt[fGW_, mchirp_] = 96 π^(8/3) (6.67 × 10^-8)^(5/3) (fGW)^(11/3) (mchirp 2 × 10^33)^(5/3) / (5 (3 × 10^10)^5)
```

Out[174]=
$$5.82578 \times 10^{-7}\ \text{fGW}^{11/3}\ \text{mchirp}^{5/3}$$

In[175]:=
```
Tsteady = 1 × 10^9 π 10^7;
```

In[176]:=
```
Integrate[dfdt[fGW, 0.5]^-1, {fGW, fmin, ∞}]
```

Out[176]=
$$\frac{2.04359 \times 10^6}{\text{fmin}^{8/3}} \quad \text{if } \text{Im}[\text{fmin}] \neq 0 \,||\, \text{Re}[\text{fmin}] > 0$$

In[177]:=
```
fmin /. Solve[2.043589184607502`*^6 / fmin^(8/3) == Tsteady, fmin][[1]]
```

Out[177]=
```
0.000151344
```

In hours, this is 3.7 hours
We basically want to reproduce the plot in Lamberts+2019. There, CO+CO is smaller by a factor around 3. HeHe and HeCO are roughly the same number.

In[178]:=
```
vobs = 1
```

Out[178]=
```
1
```

HeHe are only 7 %, corresponding to a rate of 0.0005/year

In[179]:=
```
RMWHeHe = 0.0005;
HeHefac = RMWHeHe / (π 10^7) vobs Integrate[dfdt[0.001 fGW, Mchirpf[0.3, 0.3]]^-1, fGW] 0.001 (- fGW^(8/3))
```

Out[180]=
```
9600.75
```

HeCO were 39 %, a rate of 0.003

In[181]:=
```
RMWHeCO = 0.003;
HeCOfac =
 RMWHeCO/(π 10^7) vobs Integrate[dfdt[0.001 fGW, Mchirpf[0.3, 0.8]]^-1, fGW] 0.001 (-fGW^(8/3))
```

Out[182]=
```
26 438.4
```

COCO were 54 %, a rate of 0.003

In[183]:=
```
RMWCOCO = 0.003;
COCOfac =
 RMWCOCO/(π 10^7) vobs Integrate[dfdt[0.001 fGW, Mchirpf[0.6, 0.6]]^-1, fGW] 0.001 (-fGW^(8/3))
```

Out[184]=
```
18 144.3
```

In[185]:=
```
Sumfac = (RMWHeHe + RMWHeCO + RMWCOCO)/(π 10^7) vobs
  Integrate[dfdt[0.001 fGW, Mchirpf[0.6, 0.6]]^-1, fGW] 0.001 (-fGW^(8/3))
```

Out[185]=
```
39 312.6
```

The distributions in frequency are scaled according to the merger (production) rate

In[186]:=
```
DistHeHe = ProbabilityDistribution[HeHefac z^(-8/3), {z, 0.0001, Infinity}];
DistHeCO = ProbabilityDistribution[HeCOfac z^(-8/3), {z, 0.0001, Infinity}];
DistCOCO = ProbabilityDistribution[COCOfac z^(-8/3), {z, 0.0001, Infinity}];
DistSum = ProbabilityDistribution[Sumfac z^(-8/3), {z, 0.0001, Infinity}];
pdfHeHe[z_] := PDF[DistHeHe, z];
pdfHeCO[z_] := PDF[DistHeCO, z];
pdfCOCO[z_] := PDF[DistCOCO, z];
pdfSum[z_] := PDF[DistSum, z];
```

In[194]:=
```
100 000 (vobs100pc)
```

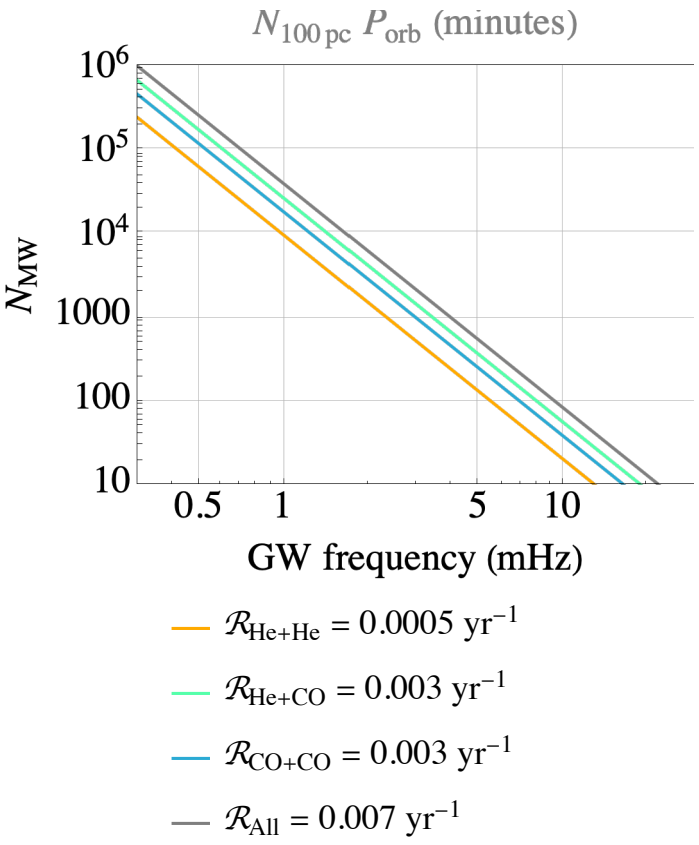
Out[194]=
```
100 000 vobs100pc
```

In[195]:=
```
Nint = {{0.001 (vobs100pc)^-1, "0.001"}, {0.01 (vobs100pc)^-1, "0.01"},
    {0.1 (vobs100pc)^-1, "0.1"}, {1 (vobs100pc)^-1, "1"}};
Nint2 = {{60 (2 / 0.001), "60"}};
```

In[197]:=

```
LogLogPlot[{pdfHeHe[z], pdfHeCO[z], pdfCOCO[z], pdfSum[z]}, {z, 0.2, 30},
 AspectRatio → 3 / 4, Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
 FrameLabel → {Style["GW frequency (mHz)"],
   Style["N_MW"], Style["N_100 pc P_orb (minutes)", Gray]},
 FrameTicks → {{Automatic, Nint}, {Automatic, Nint2}},
 FrameTicksStyle → {{Black, Gray}, {Black, Black}},
 BaseStyle → {FontSize → 20}, GridLines → Automatic,
 PlotStyle → { {Blend[{Orange, Orange, Yellow}]}, { Blend[{Cyan, Cyan, Yellow}]},
   { Blend[{Cyan, Cyan, Purple}]}, { Blend[{Black, White}]}},
 PlotLegends → {Style["ℛ_He+He = 0.0005 yr^-1", 18, Black], Style[
    "ℛ_He+CO = 0.003 yr^-1", 18, Black], Style["ℛ_CO+CO = 0.003 yr^-1 ", 18, Black],
   Style["ℛ_All = 0.007 yr^-1 ", 18, Black]},
 PlotRange → {{0.3, 30}, {10, 1000000}}]
```

Out[197]=



These curves match the population synthesis results of Lamberts+2019 Fig 7 within a factor of 2.

Now we want to estimate how many of each are in the 100 pc sample . Note that there are 5 - 10 within 100 pc in the Munday sample

In[198]:=

```
vMW = π 15^2 × 1 // N
```

Out[198]=

```
706.858
```

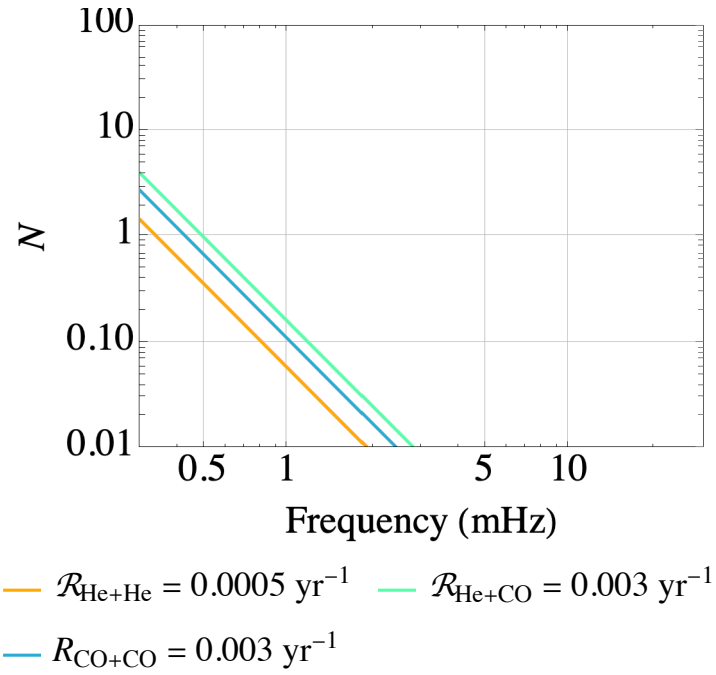
In[199]:=

```
VLr = 0.1;
```

In[200]:=
```
v100pc = 4/3 π VLr^3
```
Out[200]=
```
0.00418879
```
In[201]:=
```
vobs100pc = v100pc / vMW // N
```
Out[201]=
```
5.92593 × 10^-6
```
In[202]:=
```
DistHeHe100pc =
  ProbabilityDistribution[vobs100pc HeHefac z^(-8/3), {z, 0.0001, Infinity}];
DistHeCO100pc =
  ProbabilityDistribution[vobs100pc HeCOfac z^(-8/3), {z, 0.0001, Infinity}];
DistCOCO100pc =
  ProbabilityDistribution[vobs100pc COCOfac z^(-8/3), {z, 0.0001, Infinity}];
pdfHeHe100pc[z_] := PDF[DistHeHe100pc, z];
pdfHeCO100pc[z_] := PDF[DistHeCO100pc, z];
pdfCOCO100pc[z_] := PDF[DistCOCO100pc, z];
```
In[208]:=

In[209]:=
```
LogLogPlot[{pdfHeHe100pc[z], pdfHeCO100pc[z], pdfCOCO100pc[z]}, {z, 0.2, 30},
 AspectRatio → 3 / 4, Frame → True, LabelStyle → {(FontFamily → "Times"), Black},
 FrameLabel → {Style["Frequency (mHz)"], Style["N"]},
 BaseStyle → {FontSize → 20}, GridLines → Automatic,
 PlotStyle → { {Blend[{Orange, Orange, Yellow}]},
   { Blend[{Cyan, Cyan, Yellow}]}, { Blend[{Cyan, Cyan, Purple}]}},
 FrameTicks → {{Automatic, Automatic}, {Automatic, Automatic}},
 FrameTicksStyle → {{Black, Gray}, {Black, Black}},
 PlotLegends → {Style["ℛHe+He = 0.0005 yr^-1", 18, Black],
   Style["ℛHe+CO = 0.003 yr^-1", 18, Black],
   Style["RCO+CO = 0.003 yr^-1 ", 18, Black]}, PlotRange → {{0.3, 30}, {0.01, 100}}]
```

Out[209]=



We estimate around 50 Hubble time merger binaries in the 100 pc sample.
Torres predicts that ~ 1% of the 100 pc sample are in unresolved binaries.
The sample is ~ 10,000. So this is on the order of 1% too (0.7%)

here are the number of HeHe in the Milky Way at short orbital period:

In[210]:=
```
NHeHeMW =
 RMWHeHe / (π 10^7) vobs Integrate[dfdt[fGW, Mchirpf[0.3, 0.3]]^-1, {fGW, 0.0002, 0.5}]
```

Out[210]=
```
701819.
```

And the number in 100 pc

In[211]:=

$$\text{NHeHe100pc} = \frac{\text{RMWHeHe}}{\pi\, 10^7}\ \text{vobs100pc}\ \text{Integrate}\left[\text{dfdt}[\text{fGW}, \text{Mchirpf}[0.3, 0.3]]^{-1}, \{\text{fGW}, 0.00015, 0.5\}\right]$$

Out[211]=

8.95676

In[212]:=

$$\text{NHeCO100pc} = \frac{\text{RMWHeCO}}{\pi\, 10^7}\ \text{vobs100pc}\ \text{Integrate}\left[\text{dfdt}[\text{fGW}, \text{Mchirpf}[0.3, 0.8]]^{-1}, \{\text{fGW}, 0.00015, 0.5\}\right]$$

Out[212]=

24.665

In[213]:=

$$\text{NCOCO100pc} = \frac{\text{RMWCOCO}}{\pi\, 10^7}\ \text{vobs100pc}\ \text{Integrate}\left[\text{dfdt}[\text{fGW}, \text{Mchirpf}[0.6, 0.6]]^{-1}, \{\text{fGW}, 0.00015, 0.5\}\right]$$

Out[213]=

16.9272

So we estimate that there are 50 binaries with merger times < 1 Gyr in the 100pc sample.